C Programming

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DATA TYPES AND CONSTANTS
The four basic data types are

- **INTEGER**
  These are whole numbers, both positive and negative. Unsigned integers (positive values only) are supported. In addition, there are short and long integers.

  The keyword used to define integers is,

  ```
  int
  ```

  An example of an integer value is 32. An example of declaring an integer variable called **sum** is,

  ```
  int sum;
  sum = 20;
  ```

- **FLOATING POINT**
  These are numbers which contain fractional parts, both positive and negative. The keyword used to define float variables is,

  ```
  float
  ```

  An example of a float value is 34.12. An example of declaring a float variable called **money** is,

  ```
  float money;
  money = 0.12;
  ```

- **DOUBLE**
  These are exponetional numbers, both positive and negative. The keyword used to define double variables is,

  ```
  double
  ```

  An example of a double value is 3.0E2. An example of declaring a double variable called **big** is,

```
double big;
big = 312E+7;
```

- **CHARACTER**
  These are single characters. The keyword used to define character variables is,

  ```
  char
  ```

  An example of a character value is the letter **A**. An example of declaring a character variable called **letter** is,

  ```
  char letter;
  letter = 'A';
  ```

  Note the assignment of the character *A* to the variable *letter* is done by enclosing the value in **single quotes**. Remember the golden rule: Single character - Use single quotes.

Sample program illustrating each data type

```
#include < stdio.h >

main()
{
        int   sum;
        float money;
        char  letter;
        double pi;

        sum = 10;               /* assign integer value */
        money = 2.21;           /* assign float value */
        letter = 'A';           /* assign character value */
        pi = 2.01E6;            /* assign a double value */

        printf("value of sum = %d\n", sum );
        printf("value of money = %f\n", money );
        printf("value of letter = %c\n", letter );
        printf("value of pi = %e\n", pi );
}
```

```
Sample program output
value of sum = 10
value of money = 2.210000
value of letter = A
value of pi = 2.010000e+06
```

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